

AUDITORY GROUPING DAILY

Wednesday, 08 July 2026 · 331 papers screened · curated

Today's automated feeds were unusually noisy — the flagship-journal scan returned almost no neuroscience (plant genetics, materials chemistry, quantum computing dominated), and several Tier-3 keyword searches misfired into off-topic clinical papers, so this is a genuinely thin day. The strongest thread is prediction and boundary-detection in sequences: a Nature Communications MEG study from Németh's group on non-adjacent dependency learning, and an eLife study separating prediction-error from prediction-uncertainty event boundaries, both bearing on splitting anticipatory discrimination from representational learning in your own task. A Brody-lab bioRxiv preprint on hippocampal engrams reconfiguring prefrontal context representations is the closest thing to a precedent for context-dependent coding in prelimbic cortex. Two picks round out language/BCI: a large intelligibility jump for voice-synthesis BCIs, and a structural link between the auditory-linguistic pathway and speech-in-noise performance. Sections are kept short given the weak batch.

CHUNKING & SEQUENCE PROCESSING

Learning regularities in noise engages both neural predictive activity and representational changes.

PUBMED Coumarane Tirou et al. — *Nature Communications* — 2026

Corresponding: Romain Quentin — Université Claude Bernard Lyon 1 — France | [Open paper](#)

Using MEG during a visuomotor task, Tirou et al. show that learning non-adjacent temporal dependencies embedded in noise unfolds in two dissociable stages: fast-emerging predictive activity (stimulus-specific patterns appearing before stimulus onset, preceding any behavioral improvement) followed by a slower build-up of representational similarity between the statistically-linked, non-adjacent elements themselves. This maps onto the distinction your mouse sequence-discrimination task needs to draw between anticipatory decision-window discrimination (your d-prime measure) and slower session-level representational learning of which chords co-occur — it's a concrete two-stage hypothesis worth testing against your own early-versus-late-session data. Németh is on your watch list, and this is a direct hit on 'non-adjacent dependencies,' one of your explicitly listed core topics.

Multiple event segmentation mechanisms in the human brain.

PUBMED Tan T Nguyen et al. — *eLife* — 2026

Corresponding: Jeffrey M Zacks — Washington University in St. Louis — United States | [Open paper](#)

Nguyen et al. use fMRI plus two computational models — one driven by prediction error, one by prediction uncertainty — to show that human-perceived event boundaries in continuous experience are the product of two temporally and anatomically distinct signals: an early error-driven shift in prefrontal and anterior temporal regions, and a separate, later uncertainty-driven shift in parietal dorsal-attention regions. This is a useful formal vocabulary for your human chunk-in-stream task's distractor design: your near-miss distractors (temporal reversals, circular shifts, neighbour mutations) plausibly drive error-type boundary signals, while unrelated background chunks may lean more on the uncertainty-type signal, and this gives you a specific way to test that split in your own high-gamma or single-unit data. It's fMRI rather than intracranial, so treat the region-level claims as a coarse map to test against, not a settled circuit.

AUDITORY & PREFRONTAL ELECTROPHYSIOLOGY

Hippocampal engrams configure prefrontal context representations to guide flexible decisions

BIORXIV Julian, J. B. et al. — *bioRxiv* — 2026-07-07

Corresponding: Carlos Brody — Princeton University — United States | [Open paper](#)

Julian, Kaminsky, Tank & Brody combine hippocampal engram tagging/reactivation with large-scale recordings in mouse medial prefrontal cortex during a context-dependent task-switching paradigm, showing that reactivating a hippocampal engram for a given context reinstates that context's representation in prefrontal cortex within hundreds of milliseconds and causally redirects which stimulus-response rule the animal applies. This is a direct precedent for the kind of prelimbic-cortex, context-dependent population coding you record in your own go/no-go sequence task — it demonstrates that a fast, engram-linked context signal in prefrontal cortex can determine which decision rule governs an upcoming trial, which is exactly the kind of session-to-session context-switching your adaptive distractor-difficulty algorithm implicitly probes. Carlos Brody is on your watch list.

LANGUAGE, STROKE & BCI

Brain2voice 2.0: High-performance voice synthesis brain-computer interface

OPENALEX Maitreyee Wairagkar et al. — *bioRxiv* (Cold Spring Harbor Laboratory) — 2026-07-06

Corresponding: Sergey D. Stavisky — University of California, Davis — United States | [Open paper](#)

Wairagkar et al. (the BrainGate/UC Davis group) introduce brain2voice 2.0, a Transformer-based decoder that synthesizes real-time, intelligible voice directly from intracortical neural signals, cutting listener transcription word-error-rate from 43.75% to 5.24% — an eight-fold intelligibility improvement over the prior state of the art on the same benchmark dataset. This is squarely in your clinical/translational tier: it's a concrete picture of what a chronic intracortical array in a language-impaired participant like yours could eventually support beyond detection tasks, and their phoneme-aligned, causal real-time decoding objective is worth comparing against how you extract trial-aligned high-gamma and single-unit features in your chunk-in-stream task.

Structural connectivity of auditory-linguistic brain networks predicts success in speech categorization and listening in noise

BIORXIV Rizzi, R. et al. — *bioRxiv* — 2026-07-07

Corresponding: Gavin M. Bidelman — Indiana University Bloomington — United States | [Open paper](#)

Rizzi, Stirn, Eisenhut & Bidelman use diffusion-weighted imaging and tractography to show that individual differences in how gradiently versus discretely people categorize speech sounds, and how well they understand speech in noise, track with white-matter density in the arcuate fasciculus and brainstem-cortical auditory pathways, plus cortical thickness in right superior temporal gyrus. It's a non-invasive structural correlate rather than a mechanistic account, but it's a reminder that the statistical-learning-versus-top-down-template-matching distinction you use to frame your chunk-in-stream task has a documented individual-differences anatomical signature in the human auditory-linguistic pathway — a reason to consider collecting structural scans alongside electrophysiology in future human sessions.

OTHER NOTABLE

Development of auditory and spontaneous movement responses to music over the first postnatal year.

PUBMED Trinh Nguyen et al. — *eLife* — 2026

Corresponding: Giacomo Novembre — Italian Institute of Technology — Italy | [Open paper](#)

Nguyen et al. record EEG and markerless pose-estimated body movement simultaneously in 79 infants aged 3-12 months listening to music, finding that auditory encoding of musical structure is present from 3 months, but structured, music-driven movement (rocking, swaying, clapping-like patterns) only emerges by 12 months and even then is not beat-synchronized. It's a developmental study far from your own paradigms, but it's a clean example of your 'continuous behavior as ground truth' standing interest — treating spontaneous movement as a real, quantifiable readout of internal auditory processing rather than noise around a trial structure, in the same spirit as your GLM's slow-drift module for licking behavior.

Bregman (1990), **Auditory Scene Analysis** — foundational framework (cite a review chapter if the book isn't fetchable).

Foundations pick — abstract only

- Using MEG during a visuomotor task where participants implicitly learned non-adjacent temporal dependencies embedded in noise, the authors dissociate two temporally distinct learning mechanisms rather than treating 'statistical learning' as a single process.
- A fast, stimulus-specific predictive signal appears before stimulus onset and precedes any measurable behavioral improvement, showing the brain can encode an upcoming regularity before that knowledge shows up in behavior.
- Only afterward does a slower process build up: increased neural pattern similarity specifically between the non-adjacent elements that are statistically linked - a genuine representational reorganization of the sequence structure itself, not just a strengthening of the early predictive signal.
- Both processes are carried by a distributed set of networks including sensorimotor and dorsal attentional regions rather than one dedicated 'statistical learning area', arguing against a single, localized neural signature of sequence learning - worth keeping in mind before searching for a single region that 'does' sequence learning in prefrontal or auditory cortex.

HOW TO READ IT (≈20 min)

Full text wasn't retrievable, so budget the 20 minutes around the abstract plus whatever figures the published PDF makes available. Spend the first ~8 minutes on the Results, specifically the two analyses that produce the paper's central claim: the pre-stimulus decoding analysis that shows predictive activity emerging before behavioral improvement, and the representational-similarity analysis that shows increased pattern similarity between non-adjacent, statistically-linked elements later in learning - these are the two concrete methods worth understanding well enough to consider replicating in spike-sorted data. Spend ~5 minutes on Methods, focusing on exactly how 'non-adjacent temporal dependencies embedded in noise' were constructed in the visuomotor task, since that is the closest published methodological parallel to the mouse task's own non-adjacent generalization problem (BC = XBC = XXBC). Skim the Introduction and Discussion (~7 minutes combined), looking specifically for any claim about why the predictive signal precedes the representational one mechanistically (the abstract only asserts the ordering, not a mechanism) - if the Discussion speculates on this, that speculation is the part most worth stress-testing against prefrontal/auditory-cortex data.

YOUR QUESTION TO ANSWER

The BC-versus-CB task can only really move forward once it's known whether a mouse's licked response is riding on an early, fast anticipatory signal (paralleling this paper's stimulus-specific predictive activity) or on a slower, consolidated sequence-identity representation (paralleling their non-adjacent representational-similarity buildup) - if forced to bet which one the GLM's chunk-identity strategy weight is actually capturing at the point a mouse first reaches criterion on a new distractor stage, which would it be, and does the existing decision-window versus response-window split already allow testing this, or would a new within-session early-versus-late analysis be needed to separate them?